TrenchMobb

Project handoff: Solana meme-coin shorting site with an automatic SOL fee

Repository	https://github.com/postandstuff415/trenchmobb (branch **main**)
Owner	postandstuff415 (GitHub)
Handoff date	October 28, 2026
Stack	Vite 7 + React 19 + TypeScript, Solana wallet adapter, @solana/kit 2.3, Kamino klend-sdk 13.0.6, Jupiter Swap API
State	Code complete for v1 and builds cleanly. **Never run against a live chain**: the build environment could not reach Solana, Jupiter or Kamino.

For the Claude session reading this

You are picking up an in-progress product. Clone the repo, read this document and the README, and do not re-architect without the owner asking: the design choices below were deliberate. Your first useful jobs are probably (1) getting a working .env, (2) a live smoke test with tiny amounts, and (3) the open items in section 10. Things only the owner can supply: RPC key, fee wallet, the Kamino market, the hero artwork file, and the final spelling of the brand.

1. What the product is

A website where anyone can **short Solana meme coins**, modelled on the defunct dumpy.fun (built by Save, formerly Solend, in 2024). Each time a user opens a short, a **small flat fee in SOL** is charged automatically and sent to the owner's treasury wallet.

How a short works (spot borrow-and-sell, not perpetual futures):

- User posts **USDC collateral** to a Kamino lending market.
- User **borrows the meme token** against that collateral.
- The borrowed tokens are **sold for USDC on Jupiter**; the USDC lands in the user's wallet.
- To close: buy the tokens back with USDC, **repay** the loan, collateral is returned. If the price fell, the user keeps the difference.
- The other side is **lenders** ("bulls"): holders supply meme tokens on the Lend tab and earn the interest shorters pay.

2. Key decisions and why

Decision	Reason
Borrow + swap (Kamino + Jupiter), not perps	Matches the dumpy.fun model the owner asked for. Any token with a lending reserve can be shorted.
Not Drift Protocol	Drift is the main Solana meme-perp venue, but DefiLlama lists an April 1, 2026 exploit (~$285M, compromised admin + price manipulation). Avoided as a dependency.
Kamino over Save/marginfi	Most actively maintained SDK at build time (klend-sdk published the day of the build). Supports permissionless isolated markets, which is how meme reserves get listed.
Fee = System Program transfer inside the borrow transaction	Solana transactions are atomic: if the borrow fails, the fee is not taken. That gives "automatically deducted when they place the short" exactly.
Flat fee (default 0.01 SOL), not a percentage	Owner said "minor fee paid in SOL". Flat is simple and predictable. A bps-of-notional option is a possible follow-up.
No fee on close	Owner specified the fee on placing the short only.
Pure static frontend, no backend	Nothing custodial; cheap to host. Trade-off: the fee is enforced by the site, so someone calling Kamino directly skips it.
Vite instead of Next.js	Kamino SDK needs Node polyfills and WASM; Vite handles both with plugins more simply.
Hand-written fee and compute-budget instructions	@solana-program/system and /compute-budget require @solana/kit 3, but klend-sdk pins kit 2. The instructions are a few bytes each.

3. Transaction flows

All flows live in `src/lib/actions.ts` and go through `sendGroups()` in `src/lib/tx.ts`.

Flow	Instructions (in order)	Notes
Open short	[fee transfer] + Kamino deposit-and-borrow (or borrow only if no new collateral) + Jupiter sell meme → USDC	One atomic v0 transaction when it fits in 1232 bytes; otherwise two transactions signed together: (fee + borrow), then (sell). The fee always rides with the borrow.
Close short	Jupiter buy USDC → meme (≈ debt + 0.2%) + Kamino repay (u64::MAX = repay all) + withdraw all collateral if it was the last borrow	Needs enough USDC in the wallet; the sale proceeds from opening are kept there for this. Quote is re-scaled up to 4 times until the minimum output covers the debt.
Supply (lend)	Kamino deposit-reserve-liquidity	Lender receives kTokens (cTokens) in their wallet; no obligation account needed.
Withdraw supply	Kamino redeem-reserve-collateral for the full kToken balance	

Transaction packing details

- Compute-budget instructions from Kamino and Jupiter are stripped and replaced with one limit (1.4M CU) and one price (`VITE_PRIORITY_MICROLAMPORTS`). Duplicate compute-budget instructions would fail the tx.

- Address lookup tables from both Kamino (`action.luts`) and Jupiter are fetched and used to compress the message.
- Jupiter quotes use `maxAccounts=32` to keep routes small enough to share a transaction with Kamino.
- Kamino actions use `initUserMetadata.skipLutCreation = true` so the first short does not need a separate LUT-creation transaction.
- Wallet signs once (`signAllTransactions`); transactions are sent in order and each is confirmed before the next.
- Origination fee: Kamino adds it to the debt (verified in SDK source), so the full borrowed amount is sold.

4. Code map

File	Purpose
`src/config.ts`	Reads all VITE_* env vars; throws a readable error if required ones are missing.
`src/lib/fee.ts`	platformFeeIx(): the SOL fee transfer to the treasury.
`src/lib/instructions.ts`	Raw System transfer, SetComputeUnitLimit, SetComputeUnitPrice.
`src/lib/actions.ts`	openShort, closeShort, supplyMeme, withdrawSupply.
`src/lib/tx.ts`	Packs instruction groups into v0 tx(s), wallet signing, sending, confirmation polling.
`src/lib/jupiter.ts`	Quote + swap-instructions; converts Jupiter JSON instructions to kit Instructions.
`src/lib/kamino.ts`	RPC client, market load, collateral/shortable reserve selection, obligation PDA, token balances.
`src/lib/entries.ts`	localStorage record of sale proceeds per short, used only to display PnL.
`src/lib/format.ts`	Number formatting and Solscan links.
`src/App.tsx`	Header, tabs, hero, data loading/refresh.
`src/components/ShortPanel.tsx`	Market table + short form with max size, liquidation price, health, fee preview.
`src/components/Positions.tsx`	Open shorts, LTV vs liquidation LTV, PnL, Close button.
`src/components/LendPanel.tsx`	Supply APY, utilization, supply / withdraw-all per reserve.
`src/styles.css`	Cyberpunk theme (cyan #22e6ff / magenta #ff2bd6, Orbitron + Rajdhani fonts) matching owner's artwork.
`vite.config.ts`	React, WASM (Orca core used by Kamino SDK), Node polyfills.

5. Configuration (.env)

Copy `.env.example` to `.env`. The .env file is git-ignored.

Variable	Meaning	Default / example
`VITE_RPC_URL`	Solana RPC with getProgramAccounts (Helius, Triton, QuickNode). Public RPC will not load markets.	required

Variable	Meaning	Default / example
VITE_KLEND_MARKET	Kamino lending market holding the meme reserves.	placeholder: main market 7u3HeHxYDLhnCoErrtycNokbQYbWGzLs6JSDqGAv5PfF
VITE_COLLATERAL_MINT	Collateral / sale currency.	USDC EPjFWWd5AufqSSqeM2qN1xzybapC8G4wEGGkZwyTDt1v
VITE_SHORTABLE_MINTS	Optional comma-separated allowlist.	empty = every borrowable open-term reserve except collateral
VITE_FEE_WALLET	Treasury wallet receiving the fee.	required
VITE_FEE_SOL	Flat fee per short, in SOL.	0.01
VITE_JUP_API	Jupiter Swap API base URL.	https://lite-api.jup.ag/swap/v1
VITE_JUP_API_KEY	Key for the keyed api.jup.ag endpoint, if used.	empty
VITE_SLIPPAGE_BPS	Swap slippage.	100 (1%)
VITE_PRIORITY_MICROLAMPORTS	Priority fee per compute unit.	50000

6. Run, build, deploy

```
git clone https://github.com/postandstuff415/trenchmobb
cd trenchmobb
npm install
cp .env.example .env      # then fill it in
npm run dev               # http://localhost:5173
npm run typecheck
npm run build             # static site in dist/
```

Deploy `dist/` to any static host (Vercel, Netlify, Cloudflare Pages). Set the VITE_* variables in the host's build settings: they are baked in at build time. The build script raises Node's heap to 6 GB because the bundle is large.

7. Kamino market setup (owner task, not yet done)

The Kamino main market mostly lists majors, so the site will most likely need its own isolated market:

- Create a market (Kamino supports permissionless market creation; the SDK has a KaminoManager class for this).
- USDC reserve: collateral enabled (LTV > 0, e.g. 50–70%).
- One reserve per meme coin: LTV 0, borrowing enabled, sensible borrow limit, borrow factor > 100% for volatile coins, and a live oracle (Pyth pull/Switchboard).
- Oracles must be fresh. The app passes `scopeRefreshConfig: undefined`. If reserves use Kamino Scope feeds, a Scope refresh must be added to the flows or a crank must keep prices fresh.
- Seed liquidity / recruit lenders: short size is capped by the meme tokens supplied to each reserve.
- Put the market address in VITE_KLEND_MARKET.

8. What has and has not been verified

Verified	Not verified
TypeScript typecheck passes against klend-sdk 13.0.6 and @solana/kit 2.3 types. Production build succeeds. Page renders with no console errors at 1280px and 390px wide, with no horizontal scroll. SDK semantics checked in source: stats.loanToValue and liquidationThreshold are fractions; borrowFactor is a percent; amounts are base units; origination fee is added to debt; U64_MAX means repay-all / withdraw-all.	Any on-chain transaction (open, close, supply, withdraw). The sandbox blocked Solana RPC, Jupiter and Kamino. Whether open-short fits in one transaction for real routes (the fallback split handles it either way). Whether lite-api.jup.ag is still the right free endpoint in 2026 (configurable). Wallet compatibility beyond Wallet Standard auto-detection (Phantom, Solflare, Backpack should appear).

9. Problems hit during the build (so you don't repeat them)

- **Peer-dependency conflicts:** @solana-program/* need kit 3 (klend pins kit 2) → hand-wrote the instructions. Wallet adapter needs web3.js ≥ 1.99.
- **React types:** wallet adapter pulls @types/react 19 transitively → app moved to React 19.
- **WASM:** Kamino SDK imports Orca whirlpools core (.wasm) → added vite-plugin-wasm.
- **Out of memory:** vite-plugin-top-level-await ran out of heap on the 8 MB bundle → removed it (es2022 target supports top-level await natively).
- **Wrong USDC mint** was briefly in config; fixed to EPjFWWd5AufqSSqeM2qN1xzybapC8G4wEGGkZwyTDt1v. Double-check any hard-coded addresses.
- **Bundle size** is ~8.2 MB (1.6 MB gzipped), mostly Kamino SDK. Code-splitting is a worthwhile follow-up.
- Build warning "fs externalized" from klend-sdk signer.js is harmless in the browser (CLI-only code path).

10. Open items / next steps

#	Item	Who
1	Fill in .env: paid RPC URL, fee wallet, market address.	Owner
2	Create / configure the Kamino market with meme reserves and oracles (section 7).	Owner + Claude
3	Live smoke test with tiny amounts: supply, open short, close short, withdraw. Confirm the fee lands in the treasury and is not charged on a failed open.	Claude + owner wallet
4	Hero artwork: save the owner's cyberpunk "Trench Mob" image as public/hero.jpg (CSS already wired; neon gradient fallback until then).	Owner provides file
5	Brand spelling: the site says "TrenchMobb" (owner's instruction); the artwork says "TRENCH MOB". Confirm which.	Owner
6	Deploy dist/ to a static host with env vars set.	Claude
7	Legal: shorting/margin products are regulated in many jurisdictions (e.g. the US). Get advice; geo-block where needed.	Owner
8	Nice-to-haves: % of notional fee option; Scope oracle refresh; code-splitting; liquidation alerts; close-with-collateral (flash loan) when the wallet lacks USDC; per-short isolated obligations (today one obligation is cross-margined across all shorts).	Later

11. Known limitations and risks

- Shorts are liquidated if the token pumps enough; the UI shows estimated liquidation price and health, but there are no alerts.
- All of a user's shorts share one Kamino obligation (cross-margined): one bad short can liquidate collateral backing the others.
- Closing requires USDC in the wallet. If the user spent the sale proceeds or the token pumped, they must top up first (the UI says how much).
- Effective leverage is under 1x of collateral (borrow limited by USDC LTV ÷ meme borrow factor), similar to dumpy.fun's "small leverage".
- PnL display relies on localStorage in the browser that opened the short.
- The fee is enforced only by this frontend.

12. Suggested first prompt for the home-computer session

```
I'm continuing the TrenchMobb project (Solana meme-coin shorting site).
Repo: https://github.com/postandstuff415/trenchmobb
The handoff PDF is attached / in docs/HANDOFF.pdf: read it first.
Next: help me fill in .env, set up the Kamino market, and run a
small live test of open/close short and confirm the SOL fee.
```

Sources

DefiLlama, Drift protocol page (April 1, 2026 incident): https://defillama.com/protocol/drift?tvl=true

Yellow.com, Drift asset profile: https://yellow.com/pt/asset/drift

Bankless, How to short Solana memecoins: https://www.bankless.com/de/how-to-short-solana-memecoins

The Defiant, Dumpy.fun beta launch: https://thedefiant.io/news/defi/solana-s-dumpy-fun-launches-beta-shorting-memecoins-53-billion-market-dump

Cointelegraph, Dumpy.fun short-squeeze tool: https://cointelegraph.com/news/solana-memecoin-protocol-short-squeeze-tool

Solana Compass, Dumpy.fun: https://solanacompass.com/projects/dumpyfun

Kamino klend-sdk 13.0.6 source and type definitions (npm @kamino-finance/klend-sdk).